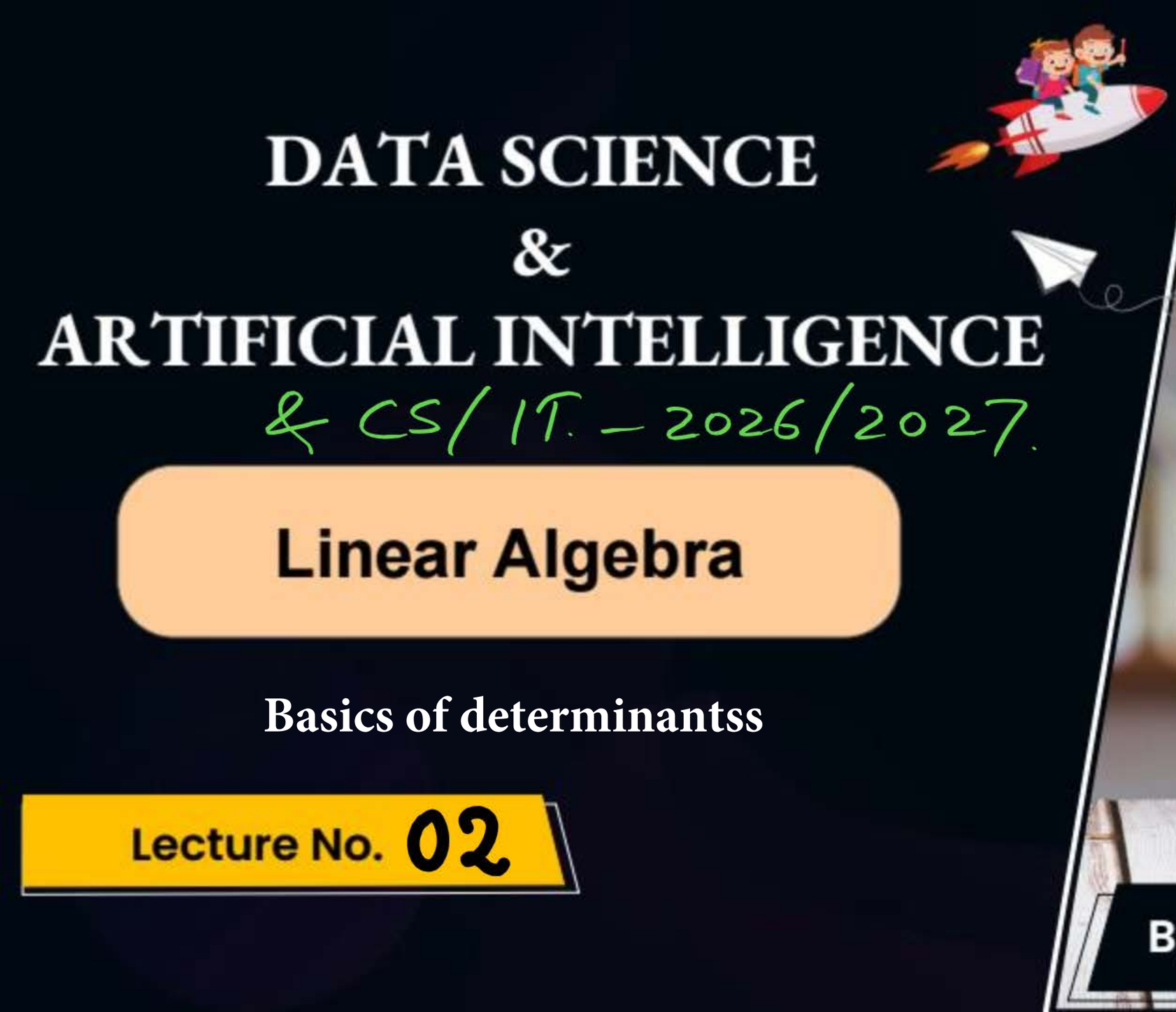
DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

Basics of determinantss

Lecture No. 02



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Recap of previous lecture



Topic BASICS of Determinants

Topics to be Covered



Topic

Properties of Determinants

& BASICS of MATRICES

Elementary Operation →

E-Row Operations

① $R_i \longleftrightarrow R_j$

② $R_i \longrightarrow kR_i$

③ $R_i \longrightarrow R_i + kR_j$

E-Column Operations

① $C_i \longleftrightarrow C_j$

② $C_i \longrightarrow kC_i$

③ $C_i \longrightarrow C_i + kC_j$

Note ① k = Any Scalar / Any Constant

② $R_3 \longleftrightarrow R_4$ ✓, $R_2 \longrightarrow 5R_2$ ✓, $R_2 \longrightarrow R_2 - 4R_3$ ✓, $R_3 \longrightarrow 5R_3 - R_2$ ✗ it is mixed operation

(3) If we interchange any two Rows in a Matrix then value of it's Det changes by −ve sign.

(4) If we multiply any Row of a Matrix by K then Value of it's Det increases by K times.

(5) If we use 3rd E-operation in a Mat, then Value of it's Det remains unaltered. i.e

we are free to apply 3rd E-operation while calculating Det.

But take care Don't use Mixed operation while Calculating Det.

Sp Note- Above Prop are also Valid for Columns.

PROPERTIES of Determinants →

① If all the elements in a Row are Zero then Det = 0

② if any two Rows in a Mat are identical then it's Det = 0

③ $|A| = |A^T|$ ④ $|A^m| = |A|^m$, $m \in I^+$ ⑤ $|ABC| = |A| \cdot |B| \cdot |C|$

⑥ $|A+B+C| = ? \neq |A|+|B|+|C|$. Calculate it Manually.

⑦ $|A^{-1}| = \frac{1}{|A|}$ or $\boxed{\det(\bar{A}') = \frac{1}{\det A}}$ where ~~$\bar{A}^{-1} = \frac{1}{A}$~~ while $\bar{A}^{-1} = \frac{\text{adj}\, A}{|A|}$

⑧ $\begin{vmatrix} a+l & b & c \\ d+m & e & f \\ g+n & h & i \end{vmatrix} = ? = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} l & b & c \\ m & e & f \\ n & h & i \end{vmatrix}$

(9) $\frac{d}{dx}\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = ? = \begin{vmatrix} a' & b' & c' \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} a & b & c \\ d' & e' & f' \\ g & h & i \end{vmatrix} + \begin{vmatrix} a & b & c \\ d & e & f \\ g' & h' & i' \end{vmatrix}$

(10) Area of $\triangle$ formed by the points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ is

$$\text{Area} = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

(11) three points A, B & C are called collinear if **Area of $\triangle ABC = 0$**

(12) we are <u>not free</u> to Multiply with any Constant in a Row (or in a Column) while solving Det But we can take out Common from any Row or from any Column.

(13) $A=\begin{bmatrix}1 & -2 & 3\\ 2 & 1 & 4\\ 0 & -1 & 2\end{bmatrix}$ & $2A=\begin{bmatrix}2 & -4 & 6\\ 4 & 2 & 8\\ 0 & -2 & 4\end{bmatrix}=2\cdot\begin{bmatrix}1 & -2 & 3\\ 2 & 1 & 4\\ 0 & -1 & 2\end{bmatrix}$

$\Rightarrow |A|=8$

$|2A|=64=8\cdot 8=2^3|A|$

ie $|2A|=\begin{vmatrix}2 & -4 & 6\\ 4 & 2 & 8\\ 0 & -2 & 4\end{vmatrix}=2\times 2\times 2\begin{vmatrix}1 & -2 & 6\\ 2 & 1 & 4\\ 0 & -1 & 2\end{vmatrix}=2^3\cdot|A|$

M. Imp

(*) if $A_{n\times n}$ & k is any scalar then $\rightarrow [kA]=k[A]$

$\rightarrow |kA|=k^n\cdot|A|$

(14) Max number of terms that can be obtained in the General expansion of $|A|_{n\times n}=n!$ terms (Learn)

eg $|A|_{2\times2} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ i.e. 2 terms obtained $= 2!$ terms.

eg $|A|_{3\times3} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\begin{vmatrix} - & - \\ - & - \end{vmatrix} - b\begin{vmatrix} - & - \\ - & - \end{vmatrix} + c\begin{vmatrix} - & - \\ - & - \end{vmatrix}$

$\cong$ (2 terms) + (2 terms) + (2 terms)

$\cong$ Max 6 terms can be obtained $= 3!$ terms.

eg $|A|_{4\times4} = \begin{vmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{vmatrix} = a\begin{vmatrix} - & - & - \\ - & - & - \\ - & - & - \end{vmatrix} - b\begin{vmatrix} - & - & - \\ - & - & - \\ - & - & - \end{vmatrix} + c\begin{vmatrix} - & - & - \\ - & - & - \\ - & - & - \end{vmatrix} - d\begin{vmatrix} - & - & - \\ - & - & - \\ - & - & - \end{vmatrix}$

$\cong$ (6 terms) + (6 terms) + (6 terms) + (6 terms)

$=$ Max 24 terms can be obtained $= 4!$ terms.

& so on - - - - -

eg: $|A| = \begin{vmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix}_{5\times5} = ?$

$R_1 \to R_1 + (R_2 + R_3 + R_4 + R_5)$

$\begin{vmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix} = 0$

eg $|A| = \begin{vmatrix} 11 & 12 & 13 & 14 \\ 21 & 22 & 23 & 24 \\ 31 & 32 & 33 & 34 \\ 41 & 42 & 43 & 44 \end{vmatrix}$

$R_4 \to R_4 - R_3$

$R_3 \to R_3 - R_2$

$R_2 \to R_2 - R_1$

$\begin{vmatrix} 11 & 12 & 13 & 14 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \end{vmatrix} = 0$

Qs if $A=\begin{bmatrix}3 & 2\\ 2 & 3\end{bmatrix}$ then $|A^3|=?$

Sol: $|A|=(3)(3)-(2)(2)=5$

So $|A^3|=|A|^3$ (By Prop 4)

$=5^3=125$

Qs if $A=\begin{bmatrix}\alpha & 2\\ 2 & \alpha\end{bmatrix}$ s.t $|A^3|=125$ then $\alpha=?$

(a) 3 (b) -3 (c) ± 3 (d) 0

$\rightarrow |A|=\alpha^2-4$ & $|A^3|=125$

$|A|=5$

$\alpha^2-4=5 \Rightarrow \alpha=\pm 3$

Qs if $A=\begin{pmatrix}1 & -\tan x & -\sec x\\ \tan x & 0 & 3\\ \sec x & 3 & \sin x\end{pmatrix}$ then $|A^T\bar{A}^{-1}|=?$

Sol: $|A^T\bar{A}^{-1}|=|A^T|\cdot|\bar{A}^{-1}|$ (By Prop 5)

$=|A|\cdot\frac{1}{|A|}$ (By Prop 3 & 7)

$=1$

Q. if $f(x)=\begin{vmatrix} x & x^2 & x^3 \\ 1 & 2 & 3x \\ 0 & 2 & 5x \end{vmatrix}$ & $f'(x)=-1$ then $x=$?

Sol: $f'(x)=\begin{vmatrix} 1 & 2x & 3x^2 \\ 1 & 2 & 3x \\ 0 & 2 & 5x \end{vmatrix}+\begin{vmatrix} x & x^2 & x^3 \\ 0 & 0 & 3 \\ 0 & 2 & 5x \end{vmatrix}+\begin{vmatrix} x & x^2 & x^3 \\ 1 & 2 & 3x \\ 0 & 0 & 5 \end{vmatrix}$

$=1[4x]-1[10x-6x^2]+x(0-6)+5(2x-x^2)$

$=-9x^2+8x$

Ans. $f'(x)=-1$ ------ $\Rightarrow x=1 \& -\frac{1}{9}$

Q. Find the area of the $\triangle$ formed by $(1,0), (2,2), (4,3)$?

Sol: $\text{Area} = \frac{1}{2}\begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{vmatrix} = \frac{1}{2}\left[2(-1)+1(-2)\right] = \frac{-3}{2}$ $\therefore \text{Area} = \frac{3}{2}$

Q. Show that $(a, b+c), (b, c+a), (c, a+b)$ are collinear.

Sol: Area of $\triangle$ formed by these points is

$$A = \frac{1}{2}\begin{vmatrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{vmatrix} \xrightarrow{C_1 \to C_1+C_2} \frac{1}{2}\begin{vmatrix} (a+b+c) & b+c & 1 \\ (a+b+c) & c+a & 1 \\ (a+b+c) & a+b & 1 \end{vmatrix}$$

$$= \frac{1}{2}(a+b+c)\begin{vmatrix} 1 & b+c & 1 \\ 1 & c+a & 1 \\ 1 & a+b & 1 \end{vmatrix} = 0 \quad (\text{By Prop 2}) \quad (\because C_1 \& C_3 \text{ are same})$$

Q: if $|A|_{3\times3} = 6$ then ① $|(2A)^{-1}| = ?$ ② $|2\cdot A^{-1}| = ?$

① $|(2A)^{-1}| = \frac{1}{|2A|} = \frac{1}{2^3|A|} = \frac{1}{8\times6} = \frac{1}{48}$

② $|2\cdot A^{-1}| = 2^3\cdot|A^{-1}| = 8\cdot\frac{1}{|A|} = 8\left(\frac{1}{6}\right) = \frac{4}{3}$

Q: if $|A|_{3\times3} = -1$, $|B|_{3\times3} = 4$ then $|5AB| = ?$

(a) 320 (b) −20

(c) −500 (d) −320

Sol: $\because A_{3\times3}\ \&\ B_{3\times3} \Rightarrow (AB)_{3\times3}$

Now $|5AB|_{3\times3} = 5^3|AB| = 125|A|.|B| = 125(-1)(4) = -500$

HWQ ① $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = ? = (a-b)(b-c)(c-a)$ Learn

HWQ ② $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = ? = (a-b)(b-c)(c-a)(a+b+c)$

CW ③ $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = ? =$

$= (a+b+c)\left[-a^2-b^2-c^2+ab+bc+ca\right]$

OR

$= -(a^3+b^3+c^3-3abc)$

OR

$= -\frac{1}{2}(a+b+c)\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$

Q(3): $|A| = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \xrightarrow{C_1 \to C_1+C_2+C_3} \begin{vmatrix} a+b+c & b & c \\ a+b+c & c & a \\ a+b+c & a & b \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$

$\xrightarrow[R_3 \to R_3-R_1]{R_2 \to R_2-R_1} (a+b+c)\begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix}$

$|A| = (a+b+c)\left[-(b-c)^2-(a-b)(a-c)\right]$

$= (a+b+c)\left[-a^2-b^2-c^2+ab+bc+ca\right]$

$= -\frac{1}{2}(a+b+c)\left[2a^2+2b^2+2c^2-2ab-2bc-2ca\right]$

$= -\frac{1}{2}(a+b+c)\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$

MATRIX

Def^n: It is the Rectangular arrangement of $m.n$ numbers. i.e

$$A=[a_{ij}]_{m\times n}=\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}$$

$1\le i\le m$

$1\le j\le n$

Horizontal lines = Row

Vertical lines = Columns

Eg $A=\begin{bmatrix} 2 & -4 & 5 \\ 1 & 2 & 7 \end{bmatrix}_{2\times 3}$ = R. Mat (Diag DNE)

Note: $A^T=A'=\begin{bmatrix} 2 & 1 \\ -4 & 2 \\ 5 & 7 \end{bmatrix}_{3\times 2}$ & $Tr(A)$ = N.D.

Sq. Mat: Def^{n} (1) No. of Rows = No of Columns or $\boxed{m=n}$

Def (2): If in a Mat Diagonal exist then it is definitely Sq. Mat.

Def (3): if $\forall$ element, $\exists$ corresponding element then Mat is Square

$$A_{m\times n} = [a_{ij}]_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}_{n\times n} = \text{Sq. Mat}$$

$1 \le i \le n$

$1 \le j \le n$

$$Tr(A) = a_{11} + a_{22} + a_{33} + \cdots + a_{nn}$$

Note ① for upper diagonal elements, $i < j$ $\forall$ i & j

② for lower " " $i > j$ " "

③ for Diagonal elements $i = j$ " "

④ Corresponding elements are a_{ij} & a_{ji}

⑤ Let $A_{n\times n}$ be any square Mat then $Tr(A)$ = Sum of Diag. elements

$= \sum a_{ii}\ \forall\ i$

✱ The Concept of Trace, Det & Inverse is defined only for Sq. Mat. while the Concept of Transpose is defined also for Rectangular Mat.

Some Special Matrices :-



U.T.M | L.T.M | Diag. Mat | Scalar Mat | Identity Mat

Trace(A) = 4 , Tr(A) = −2 , Tr(A) = 3 , Tr(A) = 20 , Tr(A) = 4

|A| = −24 , |A| = 0 , |A| = 0 , |A| = 5^4 , |A| = 1

Note:- If we want to find the Det of U.T.M, L.T.M, Diag Mat, Scalar Mat, and Identity Mat, just Multiply the Diag. elements.

i.e. **|A| = Product of Diag. elements.**

Q. if $A=[a_{ij}]_{n\times n}$ s.t $a_{ij}=0$ $\forall$ $i>j$ then A is U.T.M.

Lower Diag elements

Q. $A=\begin{bmatrix} 0 & 1 & 2 & 0 \\ -1 & 0 & 4 & 0 \\ -2 & -4 & 0 & 6 \\ 0 & 0 & -6 & 0 \end{bmatrix}$ then $|A|=?$

(a) 0 (b) 16 (c) ~~36~~ (d) -4

this Mat is Neither U.T.M nor L.T.M so we can't use the property defined on previous slide.

Hence we will try to solve it by Conventional approach

---------- Do yourself. (Ans = 36)

Analysis PODCAST

$$3 = \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix}, \begin{vmatrix} 2 & 5 \\ 1 & 4 \end{vmatrix}, \begin{vmatrix} 0 & -3 \\ 1 & 1 \end{vmatrix} \quad \text{---- } \infty \text{ ways}$$

OR

$$= \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{vmatrix} \text{ or } \begin{vmatrix} 2 & 5 & 0 \\ 0 & 0 & 1 \\ 1 & 4 & 0 \end{vmatrix} \quad \text{---- } \infty \text{ ways}$$

OR

$$= \begin{vmatrix} \cdots \\ \cdots \\ \cdots \end{vmatrix}, \quad \text{---} \quad \infty \text{ ways}$$

i.e **only Sq. Mat. have Det.** $\Rightarrow$ Det follows Basic algebraic Rules

$\because$ Det is a presentation of Real/Comp no. so

$\because$ Mat is a representation of arrangement of Nos that is why it doesn't follow Basic Algebraic Rules.

PODCAST

$$A = \begin{bmatrix} 23 & M.C & 3 & 0 & NO \\ 11 & R.C & 0 & 5 & YES \\ 5 & P.C & -1 & 2 & NO \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ _ & _ & _ & _ & _ \\ _ & _ & _ & _ & _ \end{bmatrix}$$

THANK - YOU

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